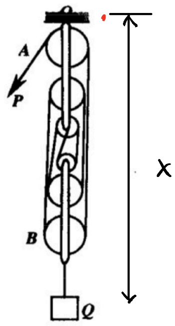
11.3.




题 11.3 图

主动力: $P$、重力 $Q$、$W$.

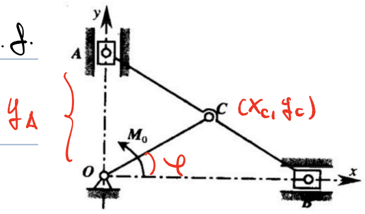
惯性力: $\frac{Q+W}{g}\cdot a$

一个自由度, 取 $x$ 为广义坐标.

$$\Rightarrow -P\cdot 6\delta x + (Q+W)\cdot \delta x - \frac{Q+W}{g}\cdot a\cdot \delta x = 0.$$

$$\Rightarrow a = \left(\frac{6P}{Q+W} - 1\right)\cdot g$$

11.8.




题 11.8 图

Lagrangian eq: $\frac{d}{dt}\frac{\partial L}{\partial \dot q_i} - \frac{\partial L}{\partial q_i} = Q_i.$

一个自由度, 取广义坐标为 $\varphi$.

$Q_\varphi$: $\sum_i Q_i \delta q_i = \sum_k \vec F_k \cdot \delta \vec r_k = \delta W$

$$Q_\varphi = \frac{\delta W}{\delta \varphi} = M_0.$$

$$L = T = \frac{1}{2}\frac{Q}{g}\dot y_A^2 + \frac{1}{2}\frac{Q}{g}\dot x_B^2 + \frac{1}{2}\left(\frac{1}{3}\frac{P}{g}a^2\right)\dot\varphi^2 + \frac{1}{2}\frac{2P}{g}(a\cdot\dot\varphi)^2 + \frac{1}{2}\left(\frac{1}{12}\frac{2P}{g}(2a)^2\right)\dot\varphi^2$$

$y_A = 2a\sin\varphi \Rightarrow \dot y_A = 2a\cos\varphi\,\dot\varphi$

$x_B = 2a\cos\varphi \Rightarrow \dot x_B = 2a\sin\varphi\,\dot\varphi$

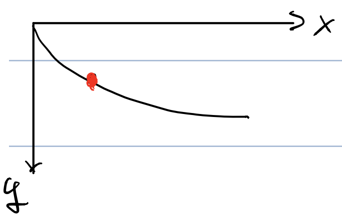
$$\Rightarrow L = \frac{1}{2}\frac{Q}{g}4a^2\dot\varphi^2 + \left(\frac{1}{6} + 1 + \frac{1}{3}\right)\frac{P}{g}a^2\dot\varphi^2$$

$$= \left(2\frac{Q}{g} + \frac{3}{2}\frac{P}{g}\right)a^2\dot\varphi^2$$

$$\Rightarrow \frac{d}{dt}\frac{\partial L}{\partial \dot\varphi} = \left(\frac{4Q}{g} + \frac{3P}{g}\right)a^2\ddot\varphi = Q_\varphi = M_0$$

$$\Rightarrow \ddot\varphi = \frac{M_0 g}{a^2(4Q+3P)}$$

11.13.




$$\begin{cases} x = R(\theta - \sin\theta) \\ y = R(1-\cos\theta) \end{cases}$$

$$T = \frac{1}{2}m(\dot x^2 + \dot y^2) = \frac{1}{2}mR^2(\dot\theta^2 - 2\cos\theta\,\dot\theta^2 + \cos^2\theta\,\dot\theta^2 + \sin^2\theta\,\dot\theta^2)$$

$$= \frac{1}{2}mR^2(1 - 2\cos\theta + 1)\dot\theta^2$$

$$= mR^2(1-\cos\theta)\dot\theta^2$$

$$V = -mgy = mgR(\cos\theta - 1)$$

$$\Rightarrow \mathcal{L} = T - V = mR^2(1-\cos\theta)\dot\theta^2 - mgR(\cos\theta - 1)$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot\theta} - \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

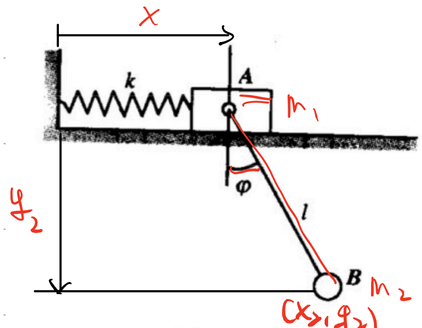
$$\Rightarrow \frac{\partial \mathcal{L}}{\partial \dot\theta} = 2mR^2(1-\cos\theta)\dot\theta \qquad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot\theta}\right) = 2mR^2\sin\theta\,\dot\theta^2 + 2mR^2(1-\cos\theta)\ddot\theta$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = mR^2\sin\theta\,\dot\theta^2 + mgR\sin\theta$$

$$\Rightarrow mR^2\sin\theta\,\dot\theta^2 + 2mR^2(1-\cos\theta)\ddot\theta = mgR\sin\theta$$

$$\sin\theta\,\dot\theta^2 + 2(1-\cos\theta)\ddot\theta = \frac{g}{R}\sin\theta.$$

11.20.




题 11.20 图

二个自由度，广义坐标取为 $x, \varphi$.

$x_2 = x + l\sin\varphi, \quad y_2 = l\cos\varphi$

$\Rightarrow \dot x_2 = \dot x + l\cos\varphi\,\dot\varphi, \quad \dot y_2 = -l\sin\varphi\,\dot\varphi$

$\Rightarrow T = \frac{1}{2}m_1\dot x^2 + \frac{1}{2}m_2(\dot x^2 + 2l\cos\varphi\,\dot x\dot\varphi + l^2\dot\varphi^2)$

$V = \frac{1}{2}kx^2 - m_2 g l\cos\varphi$

$\mathcal{L} = T - V = \cdots$

$$\frac{\partial \mathcal{L}}{\partial \dot x} = m_1\dot x + m_2\dot x + m_2 l\cos\varphi\,\dot\varphi \qquad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot x}\right) = (m_1+m_2)\ddot x - m_2 l\sin\varphi\,\dot\varphi^2 + m_2 l\cos\varphi\,\ddot\varphi$$

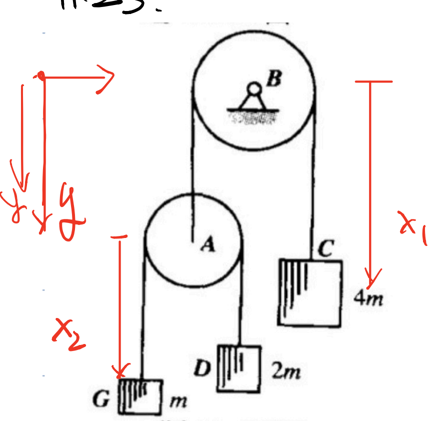
$$\frac{\partial \mathcal{L}}{\partial \dot\varphi} = m_2 l^2\dot\varphi + m_2 l\cos\varphi\,\dot x \qquad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot\varphi}\right) = m_2 l^2\ddot\varphi - m_2 l\sin\varphi\,\dot x\dot\varphi + m_2 l\cos\varphi\,\ddot x$$

$$\frac{\partial \mathcal{L}}{\partial x} = -kx \qquad \frac{\partial \mathcal{L}}{\partial \varphi} = -m_2 g l\sin\varphi$$

$$\Rightarrow \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot x}\right) - \frac{\partial \mathcal{L}}{\partial x} = 0 \Rightarrow (m_1+m_2)\ddot x - m_2 l\sin\varphi\,\dot\varphi^2 + m_2 l\cos\varphi\,\ddot\varphi = -kx$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot\varphi}\right) - \frac{\partial \mathcal{L}}{\partial \varphi} = 0 \Rightarrow m_2 l^2\ddot\varphi - m_2 l\sin\varphi\,\dot x\dot\varphi + m_2 l\cos\varphi\,\ddot x = -m_2 g l\sin\varphi$$

11.23.




题 11.23 图

$y_C = x_1 \quad y_G = l_1 - x_1 + x_2 \quad y_D = l_1 - x_1 + l_2 - x_2, \; y_A = l_1 - x_1$

$T = \frac{1}{2}4m\dot x_1^2 + \frac{1}{2}m(\dot x_2 - \dot x_1)^2 + \frac{1}{2}\cdot 2m(\dot x_1 + \dot x_2)^2 + \frac{1}{2}m\dot x_1^2$

$= m(2\dot x_1^2 + \frac{1}{2}\dot x_1^2 + \frac{1}{2}\dot x_2^2 - \dot x_1\dot x_2 + \dot x_1^2 + \dot x_2^2 + 2\dot x_1\dot x_2 + \frac{1}{2}\dot x_1^2)$

$= m(\frac{9}{2}\dot x_1^2 + \frac{3}{2}\dot x_2^2 + \dot x_1\dot x_2)$

$V = -4mgx_1 - 2mg(l_1 + l_2 - x_1 - x_2) - mg(l - x_1 + x_2) - mg(l - x_1)$

$L = T - V$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}_1} = 9m\ddot{x}_1 + m\ddot{x}_2 \qquad \frac{d}{dt}\frac{\partial L}{\partial \dot{x}_2} = 3m\ddot{x}_2 + m\ddot{x}_1$$

$$\frac{\partial L}{\partial x_1} = 4mg - 2mg - mg - mg = 0 \qquad \frac{\partial L}{\partial x_2} = -2mg + mg = -mg$$

$$\Rightarrow \quad 9m\ddot{x}_1 + m\ddot{x}_2 = 0 \qquad 3m\ddot{x}_2 + m\ddot{x}_1 = -mg$$

$-26$

$$\Rightarrow \ddot{x}_1 = \frac{1}{26}g \quad \ddot{x}_2 = -\frac{9}{26}g$$

$$\Rightarrow a_C = \ddot{x}_1 = \frac{1}{26}g \quad a_G = \ddot{x}_2 - \ddot{x}_1 = -\frac{5}{13}g \quad a_D = -\ddot{x}_1 - \ddot{x}_2 = \frac{4}{13}g$$

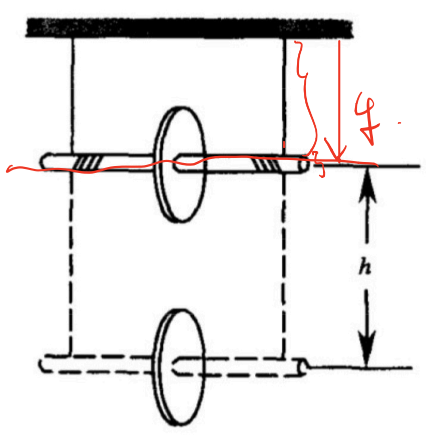
(或： $mg\,\delta y_G + 2mg\,\delta y_D + 4mg\,\delta y_C - m a_G\,\delta y_G - 2m a_D\,\delta y_D - 4m a_C\,\delta y_C$

$+ mg\,\delta y_A - m a_A\,\delta y_A = 0.$

$$(mg - m a_G)(\delta x_2 - \delta x_1) + (2mg - 2m a_D)\cdot -(\delta x_1 + \delta x_2) + (4mg - 4m a_C)\delta x_1 + (mg - m a_A)(-\delta x_1) = 0$$

$$\Rightarrow \begin{cases} -mg + m a_G - 2mg + 2m a_D + 4mg - 4m a_C - mg + m a_A = 0 \\ mg - m a_G - 2mg + 2m a_D = 0. \\ a_D + a_G = -2a_C \\ a_A = -a_C. \end{cases} \Rightarrow \begin{cases} a_G + 2a_D - 4a_C + a_A = 0. \\ 2a_D - a_G = g. \\ a_D + a_G = -2a_C \\ a_A = -a_C \end{cases} \Rightarrow \cdots$$

17.7. 取y坐标为y.




题 17.7 图

$$\omega = \frac{\dot{y}}{r}, \quad I = m\rho^2$$

$$T = \frac{1}{2}I\omega^2 + \frac{1}{2}m\dot{y}^2$$

$$= \frac{1}{2}m\dot{y}^2\left(1 + \frac{\rho^2}{r^2}\right)$$

$$V = -mgy$$

$$L = T - V = \frac{1}{2}m\dot{y}^2\left(1 + \frac{\rho^2}{r^2}\right) + mgy$$

$$p_y = \frac{\partial L}{\partial \dot{y}} = m\dot{y}\left(1 + \frac{\rho^2}{r^2}\right)$$

$$H = p_y\dot{y} - L$$

$$= \frac{1}{2}m\dot{y}^2\left(1 + \frac{\rho^2}{r^2}\right) - mgy$$

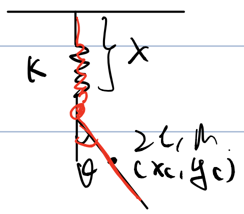
$$= \frac{1}{2}\frac{p_y^2}{m\left(1 + \frac{\rho^2}{r^2}\right)} - mgy$$

$$\begin{cases} \dot{p}_y = -\frac{\partial H}{\partial y} = mg \\ \dot{y} = \frac{\partial H}{\partial p_y} = \frac{p_y}{m\left(1 + \frac{\rho^2}{r^2}\right)} \end{cases}$$

$$\dot p_y \quad M\left(1+\frac{\rho^2}{r^2}\right)$$

$$\left( \Rightarrow \ddot{y} = \frac{g}{1+\frac{\rho^2}{r^2}} \right)$$

能量积分 $H = \frac{1}{2}\frac{p_y^2}{M(1+\frac{\rho^2}{r^2})} - Mgy$

17.13 

取广义坐标为 $x, \theta$.

$$\begin{cases} x_c = l\sin\theta \\ y_c = x + l\cos\theta \end{cases} \Rightarrow \begin{cases} \dot x_c = l\cos\theta\,\dot\theta \\ \dot y_c = \dot x - l\sin\theta\,\dot\theta \end{cases}$$

$$\begin{cases} T = \frac{1}{2}m(\dot x^2 + l^2\dot\theta^2 - 2l\sin\theta\,\dot x\dot\theta) + \frac{1}{2}\frac{1}{12}m(2l)^2\dot\theta^2 \\ \quad = \frac{1}{2}m\dot x^2 + \frac{2}{3}ml^2\dot\theta^2 - ml\sin\theta\,\dot x\dot\theta \\ V = \frac{1}{2}kx^2 - mg(x + l\cos\theta) \end{cases}$$

$L = T - V$

$$\begin{cases} p_x = \frac{\partial L}{\partial \dot x} = m\dot x - ml\sin\theta\,\dot\theta \\ p_\theta = \frac{\partial L}{\partial\dot\theta} = \frac{4}{3}ml^2\dot\theta - ml\sin\theta\,\dot x \end{cases}$$

$$\Rightarrow \dot\theta = \frac{3p_\theta + 3lp_x}{4ml^2 - 3ml^2\sin^2\theta}, \quad \dot x = \frac{3\sin\theta\,p_\theta + 4lp_x}{4ml - 3ml\sin^2\theta}.$$

$H = T + V\ (p_\theta, p_x, \theta, x)$

$$= \frac{1}{2}m\left(\frac{3\sin\theta\,p_\theta + 4lp_x}{4ml - 3ml\sin^2\theta}\right)^2 + \frac{2}{3}m\left(\frac{3p_\theta + 3lp_x}{4ml - 3ml\sin^2\theta}\right)^2 - m\sin\theta\frac{(3p_\theta + 3lp_x)\cdot(4lp_x + 3\sin\theta\,p_\theta)}{(4ml - 3ml\sin^2\theta)^2}$$

$$+ \frac{1}{2}kx^2 - mg(x + l\cos\theta)$$

$$= m(4ml - 3ml\sin^2\theta)^{-2}\cdot\Big[\frac{1}{2}(9\sin^2\theta\,p_\theta + 24\sin\theta\,l\,p_\theta p_x + 16l^2p_x^2)$$
$$+ \frac{2}{3}(9p_\theta^2 + 18lp_\theta p_x + 9l^2p_x^2)$$
$$- \sin\theta\,12lp_\theta p_x - \sin^2\theta\,9p_\theta^2 - \sin\theta\,12l^2p_x^2$$

$- \sin^2\theta \, 9 \ell P_\theta P_x]$

$P_\theta^2 (\frac{9}{2}\sin^2\theta + 6 - 9\sin^2\theta)$

$P_\theta P_x (\cancel{12\sin\theta\,\ell} + 12\ell - \cancel{12\ell\sin\theta} - 9\sin^2\theta\,\ell)$

$P_x^2 (8\ell^2 + 6\ell^2 - 12\ell^2\sin\theta)$

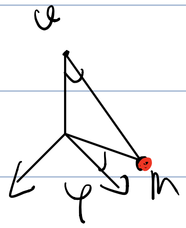
$$H = m(4m\ell - 3m\ell\sin^2\theta)^{-2} \cdot \left[ (6 - \frac{9}{2}\sin^2\theta) P_\theta^2 + (12\ell - 9\sin^2\theta\,\ell) P_\theta P_x + (14\ell^2 - 12\ell^2\sin\theta) P_x^2 \right]$$

$$+ \frac{1}{2}kx^2 - mg(x + \ell\cdot\cos\theta)$$

$$\dot{P}_x = \frac{\partial H}{\partial x} = kx - mg$$

$$\dot{P}_\theta = \frac{\partial H}{\partial \theta} = \cdots$$

17.17



$$\vec{V} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\varphi}\,\hat{\varphi}$$

$$\vec{V}^2 = \dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\varphi}^2$$

$$\begin{cases} T = \frac{1}{2}m(r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\varphi}^2) \\ V = -mgr\cos\theta \end{cases}$$

$$L = \frac{1}{2}m(r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\varphi}^2) + mgr\cos\theta$$

$$P_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} \qquad P_\varphi = mr^2\sin^2\theta\,\dot{\varphi}$$

$$H = \frac{P_\theta^2}{2mr^2} + \frac{P_\varphi^2}{2mr^2\sin^2\theta} - mgr\cos\theta.$$

$$\frac{\partial S}{\partial t} + \frac{1}{2mr^2}\left(\frac{\partial S}{\partial \theta}\right)^2 + \frac{1}{2mr^2\sin^2\theta}\left(\frac{\partial S}{\partial \varphi}\right)^2 - mgr\cos\theta = 0$$

$$S = -ht + W$$

$$\Rightarrow -h + \frac{1}{2mr^2}\left(\frac{\partial W}{\partial \theta}\right)^2 + \frac{1}{2mr^2\sin^2\theta}\left(\frac{\partial W}{\partial \varphi}\right)^2 - mgr\cos\theta = 0$$

$$W = W_1(\theta) + W_2(\varphi)$$

$$-h+\frac{1}{2mr^2}\left(\frac{d}{d\theta}W_1\right)^2+\frac{1}{2mr^2\sin^2\theta}\left(\frac{d}{d\varphi}W_2\right)^2-mgr\cos\theta=0$$

$$-2mhr^2\sin^2\theta-\sin^2\theta\, W_1'^2-2m^2gr^3\sin^2\theta\cos\theta=-W_2'^2=-C_1$$

$$\Rightarrow W_2'=\sqrt{C} \Rightarrow W_2=\sqrt{C}\varphi+C_2$$

$$C-2mhr^2\sin^2\theta-2m^2gr^3\sin^2\theta\cos\theta=\sin^2\theta\, W_1'^2$$

$$\Rightarrow \sqrt{\frac{C}{\sin^2\theta}-2mhr^2-2m^2gr^3\sin\theta\cos\theta}=\frac{dW_1}{d\theta}.$$

$$\Rightarrow W_1=\int^{\theta}d\theta\sqrt{\frac{C}{\sin^2\theta}-2mhr^2-2m^2gr^3\sin\theta\cos\theta}+C_3$$

$$S=-ht+\sqrt{C}\varphi+\int^{\theta}d\theta\sqrt{\frac{C}{\sin^2\theta}-2mhr^2-2m^2gr^3\sin\theta\cos\theta}$$

$$+C_2+C_3$$